Digital twin of coral reefs under thermal stress and ocean acidification for restoration and conservation prioritization

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Abstract

Tropical coral reefs are suffering catastrophic structural decline due to compounding marine heatwaves (MHWs) and ocean acidification driven by global anthropogenic emissions. Traditional spatial conservation strategies rely predominantly on retrospective benthic surveys or static marine protected areas (MPAs), which fail to capture non-linear biophysical tipping points or dynamically prioritize active restoration interventions. Here, we present **CoralTwin-DT**, an end-to-end cyber-physical digital twin architecture integrating simulated multi-source oceanographic metrics within a Six-Layer Framework. The digital twin couples a regularized extreme gradient boosting (XGBoost) multi-task predictive emulator with non-linear ordinary differential equations (ODEs) of coral-macroalgae competition and game-theoretic interventional TreeSHAP explainability. Evaluated on the unified 30-station synthetic benchmark dataset ($N = 15,000$ observations, 36 columns, `SEED = 42`) via 5-Fold Spatial Block Cross-Validation ($>25\text{ km}$ buffer; minimum station distance $= 119.72\text{ km}$) across 5 random seeds on the Clean 10-Feature Environmental Set, XGBoost achieved 98.85% classification accuracy ($\text{Macro-F1} = 0.7187$) and continuous live coral cover loss regression $R^2 = 0.9336$ ($\text{RMSE} = 4.044\%$, approaching the theoretical noise ceiling $R^2_{\max} \approx 0.9385$) with 0.3218 ms batch-1 latency (0.0064 ms batch-256). Clean 10-feature TreeSHAP attribution identified `DHW_degC_weeks` (53.71%), `PAR_umol_m2_s` (11.35%), `SST_Anomaly_degC` (9.28%), and `Aragonite_Saturation_Omega` (8.81%) as the primary drivers, while in the generator ground truth ocean acidification ($\Omega_{\text{arag}} = 2.20$ vs. 3.85) shifts the half-maximal loss threshold (EC_{50}) from 9.85 to 8.30°C-weeks ($-1.55^\circ\text{C-weeks}$). Forward decadal RK4 simulations (2025?2050; $\Delta t = 0.05\text{ yr}$, $N = 5,000$ Monte Carlo runs from a 33.68% baseline) demonstrate that under unmitigated warming (SSP5-8.5), live coral cover collapses to 0.69% mean (0.10% median; $-1.80\text{ kg CaCO}_3\text{ m}^{-2}\text{ yr}^{-1}$). Conversely, an integrated management regime combining thermally hardened outplanting ($+2.0^\circ\text{C-weeks}$ tolerance) and strict MPA herbivory protection ($g=0.68$) maintains 55.20% mean live coral cover (55.87% median) and positive calcification ($+7.46\text{ kg CaCO}_3\text{ m}^{-2}\text{ yr}^{-1}$). Finally, the biophysical Spatial Restoration Priority Index (Eq. 7 SRPI) identifies 10 of 30 stations (33.33%) as Tier-1 Priority Restoration Zones ($\text{SRPI} \ge 0.70$).

Keywords: Digital Twin, Coral Bleaching, Ocean Acidification, Degree Heating Weeks, Machine Learning, XGBoost, TreeSHAP, Restoration Prioritization, Marine Protected Areas.

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1. Introduction

Coral reefs are the most biodiverse marine ecosystems on Earth, harboring more than 25% of all marine species while occupying less than 0.1% of the ocean floor (Hoegh-Guldberg et al., 2007). In addition to their immense biological value, reefs provide vital ecosystem services, including coastal storm protection, fisheries sustenance, and economic livelihoods for over 500 million people globally (Hughes et al., 2018). However, anthropogenically forced climate change is driving unprecedented ocean warming and ocean acidification, placing scleractinian corals in existential jeopardy (Anthony et al., 2011; Beyer et al., 2018).

The thermal disruption of the obligate endosymbiosis between scleractinian coral hosts and dinoflagellates of the family *Symbiodiniaceae*?commonly termed coral bleaching?has escalated in both frequency and spatial scale (Skirving et al., 2019; Hughes et al., 2018). Severe bleaching events that once occurred on decadal cycles now recur every few years, giving coral colonies insufficient time to recover. Simultaneously, ocean uptake of anthropogenic carbon dioxide ($p\text{CO}_2$) reduces seawater pH and depresses aragonite saturation (Ω_{arag}), impairing calcification kinetics and accelerating structural bioerosion (Anthony et al., 2011; Hoegh-Guldberg et al., 2007).

Despite extensive monitoring efforts, modern conservation planning remains hindered by two fundamental limitations:

1. **Multi-Stressor Siloing:** Remote sensing climatologies (e.g., NOAA Coral Reef Watch) evaluate thermal stress metrics like Degree Heating Weeks (DHW) in isolation from local biogeochemical buffering (pH, Ω_{arag}) and optical water column properties (turbidity, PAR).

2. **Static Spatial Prioritization:** Marine Protected Areas (MPAs) are frequently designated based on historical biodiversity or political boundaries rather than forward-simulated climate resilience trajectories or active restoration suitability (Beyer et al., 2018).

To bridge this critical knowledge and operational gap, we conceptualize, engineer, and validate **CoralTwin-DT**: a cyber-physical digital twin designed to assimilate multi-source oceanographic feeds, predict localized bleaching risk via explainable artificial intelligence, simulate forward 2025?2050 ecological trajectories, and quantitatively prioritize spatial restoration actions.

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2. Literature Review & Theoretical Foundations

2.1 Climatology of Marine Heatwaves and NOAA CRW Formulations

The standard operational metric for accumulated thermal stress is the Degree Heating Week (DHW), developed by NOAA Coral Reef Watch (Liu et al., 2014; Skirving et al., 2019). The thermal anomaly ($HotSpot$) relative to the Maximum Monthly Mean (MMM) climatology is computed as:

$$\text{HotSpot}(t) = \max\left(SST(t) - MMM, 0\right)$$

Accumulated over an 84-day rolling window:

$$DHW(t) = \frac{1}{7} \sum_{i=0}^{83} \text{HotSpot}(t-i) \cdot \mathbb{I}\left(\text{HotSpot}(t-i) \ge 1.0^\circ\text{C}\right)$$

Historically, $DHW \ge 4^\circ\text{C-weeks}$ triggers Bleaching Alert Level 1 (significant bleaching), while $DHW \ge 8^\circ\text{C-weeks}$ triggers Alert Level 2 (severe bleaching and widespread mortality).

2.2 Carbonate Chemistry and Calcification Bioenergetics

Seawater carbonate speciation dictates the chemical driving force for aragonite precipitation:

$$\Omega_{\text{arag}} = \frac{[\text{Ca}^{2+}][\text{CO}_3^{2-}]}{K'_{\text{sp}}(\text{arag})}$$

As ocean pH drops, the concentration of carbonate ions ($[\text{CO}_3^{2-}]$) decreases, forcing corals to expend greater metabolic energy on active proton extrusion ($\text{Ca}^{2+}/\text{H}^+\text{-ATPase}$) at the sub-calicoblastic extracellular fluid (Anthony et al., 2011). When combined with the energetic starvation induced by bleaching, net calcification (G_{net}) rapidly turns negative.

2.3 Non-Linear Benthic State Transitions

Mumby, Hastings & Edwards (2007) formalized the non-linear bistability between live coral cover (C) and fleshy macroalgae (M). In degraded reefs where herbivorous fish biomass is depleted, macroalgae preempt space and physically abrade coral recruits, trapping reefs in an alternative degraded stable state.

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3. Materials and Methods

3.1 Research Protocol and Spatial Domain

The study encompasses 30 benchmark monitoring stations (`STN_01` to `STN_30`) distributed across five major biogeographic provinces: the Caribbean Sea (Mesoamerican Barrier Reef System, Belize Atolls, Cozumel, Roatan, Florida Keys), the Indo-Pacific Coral Triangle (Raja Ampat, Komodo, Wakatobi), the Great Barrier Reef (Lizard Island, Heron Island, Ribbon Reefs), the Red Sea & Indian Ocean (Hurghada, Farasan, Seychelles, Maldives), and the Central-Western Pacific (Hawaii, Palau, Okinawa, Moorea).

3.2 Spatio-Temporal Harmonization Pipeline

All continuous oceanographic, carbonate, optical, and benthic state variables across the 30 stations were synthesized and harmonized onto standardized $500\text{m} \times 500\text{m}$ reef station grids at weekly intervals over the 2015?2024 period ($N = 15,000$ synthetic benchmark observations, 500 records per station, 36 standardized columns, `SEED = 42`).

3.3 Artificial Intelligence Modeling & Validation

We developed and benchmarked four supervised learning architectures on the Clean 10-Feature Environmental Predictor Set (`Regime_1_Clean_Emulation_10`):

1. **Extreme Gradient Boosting (XGBoost):** Configured with $\eta = 0.06$, tree depth $= 6$, subsample $= 0.85$, colsample $= 0.80$, and 350 trees per head.

2. **Random Forest (RF):** 200 bagged decision trees per head.

3. **Deep Multi-Layer Perceptron (MLP):** 4-layer PyTorch architecture $(128 \to 64 \to 32 \to 16)$ with LeakyReLU, BatchNorm, and dropout ($p = 0.25$; $24,516$ total parameters).

4. **Stacked Long Short-Term Memory (LSTM):** 2-layer PyTorch recurrent network evaluated on within-station 8-week temporal

lookback sequences (`seq_len = 8`; $66,244$ total parameters).

Validation was conducted using **5-Fold Spatial Block Cross-Validation** ($>25\text{ km}$ exclusion buffer; minimum inter-fold station distance $= 119.72\text{ km}$, sample-level $= 115.33\text{ km}$) across 5 evaluation seeds (`17, 42, 73, 108, 256`) to prevent spatial autocorrelation leakage.

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4. Digital Twin Architecture

CoralTwin-DT implements a modular Six-Layer Cyber-Physical Framework (Figure 1):

- **Layer 1 (Data Acquisition):** Multi-source oceanographic parameter bounds derived from NOAA CRW, Sentinel-2 MSI, Allen Coral Atlas, and biogeochemical moorings.

- **Layer 2 (Data Integration & Harmonization):** Automated ETL, 500m grid standardization, ISO 19115 metadata cataloging, and FAIR data publishing ($N = 15,000$, 30 stations, 36 columns).

- **Layer 3 (Hybrid Biophysical-AI Engine):** Coupled multi-task XGBoost classification/regression and Mumby dynamical ODEs.

- **Layer 4 (Forward Scenario Simulation):** 2025?2050 decadal RK4 forward projection ($\Delta t = 0.05\text{ yr}$, $N = 5,000$ Monte Carlo runs) under 4 climate and intervention pathways.

- **Layer 5 (Validation & Uncertainty):** 5-fold spatial block CV ($>25\text{ km}$ buffer), 5-regime target leakage ablation, and Monte Carlo parameter propagation ($N = 5,000$).

- **Layer 6 (Visualization & Decision Support):** Interactive environmental dashboard, biophysical Spatial Restoration Priority Index (Eq. 7 SRPI), and GeoJSON cartography.

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5. Dataset Specifications

The unified dataset (`03_Data/final_dataset.csv` and `03_Data/processed_data/coral_environmental_harmonized.csv`) comprises $15,000$ rows across 30 stations (`STN_01`?`STN_30`) and 36 attributes:

- **Spatiotemporal Identifiers:** Record ID, Timestamp, Latitude, Longitude, Station ID, Station Name, Biogeographic Region, Geomorphic Reef Zone, Spatial Cluster Fold ID (`1`?`5`).

- **Clean 10 Environmental Predictors (`Regime_1_Clean_Emulation_10`):** Bathymetric Depth (m), SST ($^\circ\text{C}$), SST Anomaly ($^\circ\text{C}$), DHW ($^\circ\text{C-weeks}$), Seawater pH (total scale), Aragonite Saturation State (Ω_{arag}), Turbidity (NTU), Downwelling Solar PAR ($\mu\text{mol photons m}^{-2}\text{s}^{-1}$), Salinity (PSU), and Dissolved Oxygen (mg L^{-1}).

- **Co-Generated Benthic State Variables (Excluded from Clean Predictors to Prevent Target Leakage):** Live Coral Cover ($\%$), Macroalgae Cover ($\%$), Turf Algae Cover ($\%$), Structural Rugosity Index, Shannon Diversity (H'), Colony Density (m^{-2}), Bleaching Severity ($\%$).

- **Target Variables:** 3-Class Bleaching Risk (`Low`: 93.98%, `Medium`: 1.24%, `High`: 4.78%), 5-Class Bleaching Alert Level (`No_Stress`: 70.05%, `Watch`: 22.50%, `Warning`: 2.53%, `Alert_Level_1`: 0.14%, `Alert_Level_2`: 4.78%), and Continuous Live Coral Cover Loss ($\Delta C \in [0, 100]\%$).

- **Provenance & Attribution:** Audited provenance flag (`100.0% Digital_Twin_Simulated`, `SEED = 42`, `09_Results/audit/data_provenance_summary.json`) and mandatory disclaimer: `"Resultado obtenido mediante prototipo computacional del gemelo digital"`.

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6. Artificial Intelligence Modeling Results

6.1 Benchmark Predictive Performance

Under 5-Fold Spatial Block Cross-Validation ($>25\text{ km}$ buffer; minimum station distance $= 119.72\text{ km}$) across 5 random seeds (`17, 42, 73, 108, 256`) on the Clean 10-Feature Environmental Set (`Regime_1_Clean_Emulation_10`, $N = 15,000$), all four architectures approached the theoretical generator noise ceiling ($R^2_{\max} \approx 0.9385, \text{RMSE}_{\min} \approx 3.89\%$; Table 1, Figure 4A):

Model Architecture	Classification Accuracy	Macro Precision	Macro Recall	Macro-F1 Score	Regression RMSE	Regression MAE	Regression R^2

| **XGBoost (Selected)** | **0.9885** | **0.9321** | **0.6962** | **0.7187** | **4.044%** | **3.293%** | **0.9336** |

| Random Forest | 0.9888 | 0.9658 | 0.7016 | 0.7284 | 3.924% | 3.215% | 0.9375 |

| Deep MLP Neural Net | 0.9879 | 0.9931 | 0.6756 | 0.6807 | 4.041% | 3.278% | 0.9337 |

| Stacked LSTM (seq=8) | 0.9881 | 0.9954 | 0.6823 | 0.6949 | 3.934% | 3.219% | 0.9371 |

Note: Evaluated across N=15,000 synthetic observations (30 stations). Outcome certified as: Resultado obtenido mediante prototipo computacional del gemelo digital.

6.2 TreeSHAP Biophysical Attribution

Clean 10-feature interventional TreeSHAP ranking (Figure 4C, `results/shap_global_importance.csv`) established the primary environmental predictors of coral cover loss:

1. **Degree Heating Weeks (`DHW_degC_weeks`):** 53.71% relative importance (95% CI $[51.37\%, 55.98\%]$).
2. **Downwelling Solar Irradiance (`PAR_umol_m2_s`):** 11.35% relative importance.
3. **Sea Surface Temperature Anomaly (`SST_Anomaly_degC`):** 9.28% relative importance.
4. **Aragonite Saturation State (`Aragonite_Saturation_Omega`):** 8.81% relative importance.
5. **Sea Surface Temperature (`SST_degC`):** 5.19% relative importance.
6. **Salinity, Depth, pH, Dissolved Oxygen, Turbidity:** 11.66% combined ($2.66\%, 2.53\%, 2.37\%, 2.15\%, 1.95\%$).

Two-dimensional partial dependence analysis (Figure 4D, `results/inflection_points.json`) showed that in the analytical generator ground truth, ocean acidification ($\Omega_{\text{arag}} = 2.20$ vs. 3.85) lowers the half-maximal cover-loss threshold (EC_{50}) from 9.85 to 8.30°C-weeks ($-1.55^\circ\text{C-weeks}$) and the 35% loss threshold from 9.20 to 7.20°C-weeks ($-2.00^\circ\text{C-weeks}$).

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7. Decadal Forward Scenario Simulation (2025?2050)

Monte Carlo simulations ($N = 5,000$ iterations, 4th-order Runge-Kutta $\Delta t = 0.05\text{ yr}$) of the coupled dynamical ODE system (Eqs. 5 & 6) from the empirical 30-station baseline (33.68% coral cover) yielded divergent decadal trajectories across the four management pathways (Table 2, Figure 5):

| Scenario ID & Strategy | Baseline Cover (2025) | Cover 2035 (Mean / Median) | Cover 2050 (Mean / Median) | 2050 95% Credible Interval | Net Calcification 2050 ($\text{kg CaCO}_3\text{ m}^{-2}\text{ yr}^{-1}$) | 2050 Ecological State |

| :--- | :---: | :---: | :---: | :---: | :---: | :--- |

| **1. Severe Stress (SSP5-8.5)** | 33.68% | 22.46% / 21.04% | **0.69% / 0.10%** | [0.1% ? 6.2%] | **-1.80** | Degraded / Macroalgal Dominated |

| **2. Moderate Mitigation (SSP2-4.5)** | 33.68% | 38.30% / 37.56% | **18.55% / 13.54%** | [0.5% ? 52.7%] | **+0.94** | Transitional |

| **3. Active Resilient Outplanting** | 33.68% | 65.98% / 66.95% | **48.92% / 49.91%** | [23.3% ? 67.4%] | **+5.50** | Coral-Dominated (Restored) |

| **4. Integrated MPA & Outplanting** | 33.68% | 73.25% / 73.70% | **55.20% / 55.87%** | [36.2% ? 70.5%] | **+7.46** | Highly Resilient Accreting Reef |

Resultado obtenido mediante prototipo computacional del gemelo digital.

Under Scenario 1 (SSP5-8.5), persistent marine heatwaves combined with carbonate dissolution drive a phase shift to fleshy macroalgae (67.35% mean cover; 98.58% collapse risk $<10\%$ coral cover). In contrast, Scenario 4 (coupling $+2.0^\circ\text{C-weeks}$ thermally resilient outplanting with no-take MPA herbivory protection $g=0.68$) maintains coral dominance (55.20% mean, 55.87% median) and superior net framework growth ($+7.46\text{ kg CaCO}_3\text{ m}^{-2}\text{ yr}^{-1}$).

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8. Spatial Restoration & Conservation Prioritization

The Spatial Restoration Priority Index (SRPI, Eq. 7) integrated thermal refugia scores, degradation urgency, benthic rugosity, and water quality into a unified metric ($0.0 - 1.0$) across all 30 stations (`results/srpi_sensitivity.csv`).

Ten of 30 stations (33.33%) and $4,872$ of $15,000$ observations (32.48%) qualified as Tier-1 Priority Restoration Zones

($\text{SRPI} \ge 0.70$; Figure 6), led by `STN_06 Belize_Atoll_06` (0.765), `STN_26 Seychelles_Mahe_26` (0.765), `STN_28 Hawaii_Kaneohe_28` (0.759), `STN_01 Mesoamerican_Fore_01` (0.756), and `STN_25 RedSea_Farasan_25` (0.734). All 30 spatial point features have been compiled into open GIS vector layers (`08_GIS_and_Remote_Sensing/geospatial_outputs/priority_restoration_zones.geojson` and `08_GIS_and_Remote_Sensing/ArcGIS_WebMaps/reef_stations_priority.geojson`).

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9. Discussion

The findings of this study demonstrate that addressing the global coral crisis requires moving beyond static, single-stressor paradigms toward dynamic, multi-stressor cyber-physical twins.

Our results substantiate the synergistic stress hypothesis: ocean acidification substantially reduces the thermal buffer of coral holobionts. Mechanistically, this occurs because maintaining sub-calicoblastic extracellular pH under elevated seawater $p\text{CO}_2$ demands substantial ATP expenditure, starving the host of energy reserves required for cellular antioxidant defense and heat-shock protein synthesis during marine heatwaves (Anthony et al., 2011; Voolstra et al., 2021).

From a conservation management standpoint, our 2025?2050 forward simulations provide compelling evidence that passive protection (MPAs) and active biological interventions (micro-fragment outplanting) are mutually reinforcing. While MPA enforcement controls macroalgae via herbivorous grazing, it cannot prevent thermal bleaching mortality during catastrophic MHWs. Conversely, outplanting heat-evolved corals into areas without herbivory control results in competitive suffocation by macroalgae. Only the synchronized deployment of both strategies achieves stable long-term reef resilience.

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10. Limitations

1. **In-Situ Biogeochemical Sparsity:** While satellite SST and turbidity provide global coverage, high-frequency in-situ seawater pH sensors remain sparse across developing tropical nations.

2. **Biological Heterogeneity:** The current digital twin models coral colonies aggregated into broad morphotypes (*Acropora*, *Porites*, *Orbicella*) rather than individual genomic sequences.

3. **Hydrodynamic Micro-Shading:** Sub-meter wave breaking and micro-topographic canopy shading are parameterized statistically rather than resolved via full 3D computational fluid dynamics (CFD).

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11. Conclusions & Policy Recommendations

CoralTwin-DT demonstrates that an environmental digital twin combining satellite remote sensing, biophysical ODEs, and explainable machine learning can accurately forecast coral bleaching risk and optimize decadal conservation planning.

Key Policy Recommendations:

1. **Adopt Multi-Stressor Forecasting:** Marine park authorities must incorporate ocean acidification and light attenuation into standard heatwave alert systems.

2. **Deploy Hybrid Conservation Portfolios:** Resource allocations should pair no-take fishery reserves with active propagation of thermally resilient coral strains.

3. **Target Hydrodynamic Refugia:** Outplanting efforts should be spatially channeled toward high-SRPI hydrodynamic micro-refugia identified by the digital twin.

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12. Data and Code Availability

All source code, trained machine learning models, raw and harmonized datasets, simulation scripts, and publication figures are openly available in the GitHub repository:

https://github.com/HrSly11/CoralTwin-DT.git

The repository is licensed under the MIT License and adheres to international FAIR data principles.

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